137 2425 PSA - Practice for unit 1 and 2 for midt term

Calculus (University of Toronto)



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MAT 137
Problem Set A

This problem set is intended to help you prepare for Test 1. It is not comprehensive: it only contains some problems that were not included in past problem sets or in past tutorials. You do not need to turn in any of these problems.

For further preparation review pre-class quizzes, lecture notes, Problem Set 1, Tutorials 1 through 3, and Unit 1 and 2 practice problems.

Problems

1. Let f be a function. Let $a, L \in \mathbb{R}$. Assume that f is defined on some open interval around a, except maybe at a. As you know, the definition of the statement $\lim\limits_{x\to a} f(x) = L$ is

   (1) For every $\varepsilon > 0$, there exists $\delta > 0$ such that $\quad 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

   Below is a list of several statements.

   (a) For every $\varepsilon > 0$, there exists $\delta > 0$ such that $\quad 0 < |x - a| < \delta \implies 0 < |f(x) - L| < \varepsilon$. Compare this statement with the definition of $\lim\limits_{x\to a} f(x) = L$. What does this statement mean? Hint: Can you find a function $f(x)$ that satisfies this statement? Compare $f(x) = x$ and $f(x) = x^2 \sin(\frac{1}{x})$. Use Desmos to graph these two functions. (https://www.desmos.com/calculator) Do these two functions satisfy this statement?

   (b) For every $\varepsilon \geq 0$, there exists $\delta > 0$ such that $\quad 0 < |x-a| < \delta \implies |f(x)-L| < \varepsilon$. Can you find a function satisfy this statement? If yes, give one example. If no, can you explain why. Hint: can you write down the negation of this statement? The negation is true or false?

   (c) For every $\varepsilon > 0$, there exists $\delta > 0$ such that $\quad |x - a| < \delta \implies |f(x) - L| < \varepsilon$. Compare this statement with the definition of $\lim\limits_{x\to a} f(x) = L$. What's the difference? What does this statement mean? Hint: sketch the graphs of the functions $f(x) = x$ and $f(x) = \begin{cases} x, & x \neq 1 \\ 2, & x = 1 \end{cases}$. Let $a = 1$ and $L = 1$. Check if they are satisfies (1) and (d).

   (d) For every $\varepsilon > 0$, there exists $\delta \geq 0$ such that $\quad 0 < |x-a| < \delta \implies |f(x)-L| < \varepsilon$. Can you find a function satisfies this statement? If yes, give one example. If no, can you explain why. Hint: what happens if I take $\delta = 0$?

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(e) There exists $\delta > 0$ such that for every $\varepsilon > 0$, $\quad 0 < |x-a| < \delta \implies |f(x)-L| < \varepsilon$. What does this statement mean?

2. Prove the following version of the Squeeze Theorem:

Theorem. *Let f and g be functions with domain $\mathbb{R}$. IF*

- *for all x large enough, $f(x) \leq g(x)$*
- $\lim_{x\to\infty} f(x) = \infty$

THEN $\lim_{x\to\infty} g(x) = \infty$

3. Sketch the graph of a function g defined on $(-5, 5)$which satisfies the following properties at once:

- $\lim_{x\to 2} g(x) = 3$
- $\lim_{x\to 2} g(g(x)) = -1$
- $\lim_{x\to -2} g(x)$ does not exist
- $\lim_{x\to -2} [g(x)]^2$ exists
- g has a removable discontinuity at $x = 0$.

4. Sketch the graph of a function f defined on $\mathbb{R}$ which satisfies the following properties at once.

- $\lim_{x\to 0} f(x) = -2$
- $\lim_{x\to 0} f(f(x)) = 1$
- $\lim_{x\to 0^-} f(\frac{1}{x}) = 3$
- $\lim_{x\to 2^+} f(x) = -\infty$
- $\lim_{x\to 2} [f(x)]^2 = \infty$
- $\lim_{x\to 3} f(x) = -1, f(3) = 2$

5. Let f be a function defined on an interval centered at 4.

(a) Write the formal ϵ-δ definition of " f is continuous at 4".

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(b) Assume f is continuous at 4 and $f(4) > 0$. Prove that there exists an open interval I, centered at 4, such that

$$\forall x \in I, f(x) > \frac{f(4)}{2}$$

Write a proof directly from the above formal ϵ-δ definition of continuity. Do not use any of the limit laws or any other theorems.

6. Let $f(x)$ be a function defined on $\mathbb{R}$ such that $\lim_{x\to\infty} f(x) = 2$. Use the formal ε-δ definition of the limit to prove

$$\lim_{x\to\infty} 4f(3x+1) = 8.$$

Do not use any limit laws or any other theorems.

7. Prove the following theorem:

Theorem. *Let $a \in \mathbb{R}$ and let f be a function defined on an interval centered at a. IF f is continuous at a and $f(a) > 0$ THEN f is strictly positive on some interval centered at a.*

8. Prove the $\lim_{x\to 0^+} \frac{1}{x}$ does not exist from the formal $\varepsilon - \delta$ definition.
Note that the formal definition of $\lim_{x\to a} f(x)$ does not exist and the definition of $\lim_{x\to a} f(x) = \infty$ are different.

9. Let $a \in \mathbb{R}$ and let f be a function with domain $\mathbb{R}$. Using the formal definition of the limit, prove that:

IF $\lim_{x\to a} f(x) = \infty$ THEN $\lim_{x\to a} f(x)$ does not exist.

10. Define a sequence $\{x_n\}$ where $n \in \mathbb{Z}$ and $n \geq 0$ by the following rule:

$$a_0 = 1, a_1 = 3, a_2 = 9, a_n = a_{n-1} + 3a_{n-2} + 9a_{n-3} \text{ for } n \geq 3.$$

Prove that $a_n = 3^n$ for all $n \in \mathbb{Z}$ and $n \geq 0$. Hint: use the induction method.

11. For every positive number $x > 0$ and for every natural number $n \geq 2$,

$$(1+x)^n > 1 + nx.$$

Hint: Use induction.

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12. Define $f(n) = 1^2 + 2^2 + 3^2 + \cdots + n^2$ where n is any positive integer.
Prove the following statement by using math induction.

$$P(n) : f(n) \leq n^3$$

is true for all positive integer n.

13. Let A, B be any subsets of $\mathbb{R}$ and non-empty.
We say A is *calm* if

$$\exists a, b \in A, \ \ s.t. \ \forall x \in A, a \leq x \leq b$$

We say B is *happy* if

$$\exists a \in B, \ \ s.t. \ \forall x \in B, x \geq a$$

Determine if the following statements are true or false. If a claim is true, prove it. If a claim is false, provide a counterexample and a justification of how the counterexample shows the claim is false.

(a) If a non-empty set B is happy, then it is calm.

(b) If a non-empty set A is calm and a non-empty set B is happy, $A \neq B$, then $A \cup B$ is happy.

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